Working through the differences between IaaS, PaaS and SaaS this week really clarified something that always felt a bit abstract to me before, which is that the three models are not just different products, they represent different amounts of control an organisation is willing to give up in exchange for convenience. IaaS sits closest to traditional infrastructure, giving access to raw compute, storage and networking while leaving the operating system, runtime and applications entirely up to the business, which is why it tends to suit organisations that want cloud scalability but still need full control over their environment, AWS EC2 and Azure Virtual Machines being the obvious examples. PaaS moves further up the stack by managing the underlying infrastructure and runtime, so developers can focus purely on building and deploying applications, which is where platforms like Azure App Service or Google App Engine fit in. SaaS goes the furthest, delivering a finished application that users simply consume, with Microsoft 365 and Salesforce being the everyday examples most people already use without necessarily calling it cloud computing.

What stood out most to me from looking at current market data is that SaaS still holds the largest share of the overall cloud market, but IaaS is actually growing the fastest right now, driven largely by organisations shifting away from legacy on premises infrastructure and by the sheer compute demand that AI and machine learning workloads are creating (Grand View Research, 2026). That timing makes sense to me, since training and running modern AI models needs the kind of elastic, high performance infrastructure that only IaaS can really provide at scale. Reflecting on this alongside the original NIST service model definitions (Mell and Grance, 2011), it is clearer to me now that choosing between these models is really a question of where an organisation wants to spend its effort, on infrastructure, on development, or on neither.

References

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